Homework #1 Solutions

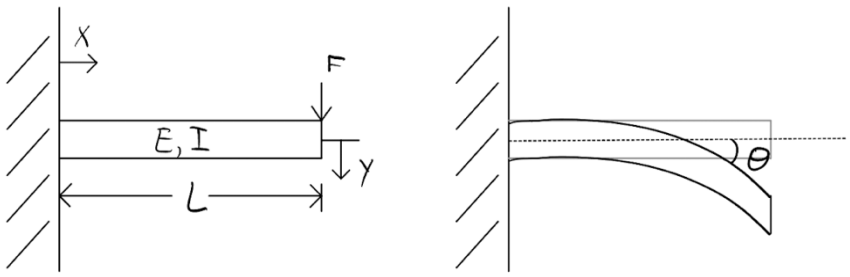
1) For hydraulic systems, the linear capacitance, C_f, is given by

$$C_f = \frac{\forall}{P}$$

Where ∀ is the fluid volume and P is the hydrostatic pressure. ∀ and P are given by $\forall = Ah$ and $P = \rho g h$, respectively. Substituting ∀ and P into the first equation gives

$$\boxed{C_f = \frac{Ah}{\rho g h} = \frac{A}{\rho g}}$$

2) A cantilever beam can be modeled as a torsional spring or translational spring. Either is correct.



Note: $\theta = f(x), y = f(x)$

For a **<u>torsional</u>** spring:

The differential equations governing the angular and vertical deflection of the beam as a function of the distance along the beam are given by

$$\theta = \frac{dx}{dy} \qquad \text{and} \qquad \frac{M}{EI} = \frac{d^2y}{dx^2}$$

Where M is the moment which can be expressed as M = FL

First, note that

$$\theta = \frac{dx}{dy} => \frac{d\theta}{dx} = \frac{d^2y}{dx^2} = \frac{M}{EI}$$

For a torsional spring, we can consider the angular deflection at the end of the beam and relate it to the moment applied to get the capacitance, i.e., $C_R = \frac{\theta}{M}$ if the relationship is linear. To solve for θ, we have:

$$\frac{d\theta}{dx} = \frac{M}{EI} => \int \frac{M}{EI} dx$$

$$\theta(x) = \frac{M}{EI}x + C$$

Using boundary condition, $\theta(x = 0) = 0$, we get C = 0 and

$$\theta(x) = \frac{M}{EI}x$$

At x = L,

$$\theta(L) = \frac{ML}{EI}$$

Rearranging gives

$$\boxed{C_R = \frac{\theta}{M} = \frac{L}{EI}}$$

For a **<u>translational</u>** spring:

Starting at the boundary conditions of the previous method…

$$\theta(x) = \frac{M}{EI}x$$

$$M = Fx$$

Therefore,

$$\frac{dy}{dx} = \frac{Fx^2}{EI}$$

To solve for y, we have:

$$\int dy = \int \frac{F}{EI}x^2 dx$$

$$y = \frac{F}{EI}\frac{x^3}{3}$$

At x = L,

$$y(L) = \frac{FL^3}{3EI}$$

Rearranging gives

$$\boxed{C_T = \frac{y}{F} = \frac{L^3}{3EI}}$$

3) We consider the fluid in the pipe as a rigid body. Newton's Law is given by

$$F = ma = \frac{dp}{dt} = \frac{d(mv)}{dt}$$

If F_2 is the net force acting on the fluid by the difference in pressure $P_2 = P_1 - P_3$, then F_2 can be written as

$$F_2 = P_2 A = \frac{d(mv_2)}{dt} = \frac{d(\rho A L v_2)}{dt}$$

Where v_2 is the average velocity of the fluid. v_2 can be expressed as a function of the volumetric flowrate, Q_2, as $v_2 = Q_2/A$. Then, we have

$$P_2 = \frac{1}{A}\frac{d(\rho A L\left(\frac{Q_2}{A}\right))}{dt} = \frac{\rho L}{A}\frac{dQ_2}{dt}$$

We can now substitute P_2 to verify the relationship between the pressure momentum Γ_2 and volume flow, Q_2.

$$\boxed{\Gamma_2 = \int^t \frac{\rho L}{A}\frac{dQ_2}{dt}dt = \frac{\rho L}{A}Q_2}$$